

Vacuum Minkowski Local Quantum Theory

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1 Killing Vectors & Conserved Quantities

Along a geodesic,

$$U^\mu = \frac{dx^\mu}{d\lambda}.$$

If V^μ is a Killing vector, then

$$V_\mu U^\mu$$

is conserved along the geodesic.

Indeed,

$$U^\nu \nabla_\nu (V_\mu U^\mu) = U^\nu U^\mu \nabla_\nu V_\mu + V_\mu U^\nu \nabla_\nu U^\mu.$$

Since

$$U^\nu \nabla_\nu U^\mu = 0$$

along a geodesic, and

$$\nabla_{(\mu} V_{\nu)} = 0$$

for a Killing vector, the quantity is conserved.

If the metric does not depend on an angle coordinate, then the corresponding angular momentum is conserved.

For an n -dimensional manifold, the maximum number of independent Killing vectors is

$$\boxed{\frac{n(n+1)}{2}}.$$

They consist of

n translations

and

$$\frac{n(n-1)}{2}$$

rotations.

A spacetime with the maximum number of Killing vectors is called

maximally symmetric.

It is both

homogeneous and isotropic.

Example: $\mathbb{R}^2$

$$ds^2 = dx^2 + dy^2,$$

with

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

which is flat.

There are

2 translations

and

1 rotation.

By definition,

$$\mathcal{L}_V g_{\mu\nu} = 0,$$

which defines a Killing vector.

Thus,

$$\mathbb{R}^2$$

is maximally symmetric.

Example: $\mathbb{R}^n$

Flat Euclidean space

$$ds^2 = (dx^1)^2 + \cdots + (dx^n)^2.$$

There are

$$n$$

translation generators,
and

$$\frac{n(n-1)}{2}$$

rotation generators.

Hence,

$$n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$$

independent Killing vectors.

Example: Minkowski spacetime

$$\mathbb{R}^{1,3},$$

Lorentzian flat spacetime,

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2,$$

with

$$R^\mu{}_{\nu\rho\sigma} = 0.$$

Since $n = 4$,

$$\boxed{\frac{4(4+1)}{2} = 10}$$

independent Killing vectors.

Therefore Minkowski spacetime is maximally symmetric.

2 Poincaré Group

Minkowski spacetime has

$$10$$

symmetries, forming the Poincaré group.

1. Spacetime translations

There are four translations,

$$x^\mu \rightarrow x^\mu + a^\mu,$$

whose generators are

$$P_\mu = \partial_\mu.$$

Translation symmetry implies

Energy-momentum conservation.

2. Lorentz transformations

There are six Lorentz transformations.

1. Three spatial rotations.

For example,

$$\begin{aligned}x' &= x \cos \theta - y \sin \theta, \\y' &= x \sin \theta + y \cos \theta,\end{aligned}$$

which leave

$$dx^2 + dy^2 + dz^2$$

invariant.

Generators:

$$J_x, \quad J_y, \quad J_z.$$

Corresponding conserved quantity:

Angular momentum.

2. Three boosts.

For an x -boost,

$$\begin{aligned}t' &= \gamma(t - vx), \\x' &= \gamma(x - vt), \\y' &= y, \\z' &= z,\end{aligned}$$

where

$$\gamma = \frac{1}{\sqrt{1 - v^2}}, \quad (c = 1).$$

This keeps

$$ds^2$$

invariant.

Generators:

$$K_x, \quad K_y, \quad K_z.$$

Therefore,

Poincaré group = Lorentz group + spacetime translations

The Lorentz group keeps the spacetime origin fixed.

For example,

$$SO(2)$$

is the rotation group in $\mathbb{R}^2$,
while

$$ISO(2)$$

is the Euclidean group.

Similarly,

$$O(1,3)$$

preserves the Lorentzian metric,

$$\Lambda^T \eta \Lambda = \eta.$$

The inhomogeneous Lorentz group is

$$ISO(1,3).$$

All ten generators of the Poincaré group correspond to Killing vectors of Minkowski spacetime.

Since

$$\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$$

is constant,

$$\mathcal{L}_V \eta_{\mu\nu} = 0.$$

Hence Minkowski spacetime possesses the maximum number of Killing vectors.

3 Spacetime & Operator Conventions

The Killing vectors of Minkowski spacetime can be written as

$$\textbf{Translations:} \quad \xi^\mu = a^\mu,$$

whose generators are

$$\partial_t, \quad \partial_x, \quad \partial_y, \quad \partial_z.$$

Lorentz transformations:

$$\xi^\mu = \omega^\mu{}_\nu x^\nu,$$

where

$$\omega_{\mu\nu} = -\omega_{\nu\mu},$$

so that $\omega_{\mu\nu}$ is antisymmetric and has 6 degrees of freedom.
Hence there are in total 10 independent Killing vectors.

A vector

$$x \in M^D$$

is spacelike when

$$x^2 := \eta_{\mu\nu} x^\mu x^\nu > 0.$$

(Lorentzian signature convention.)

A test function is

$$f \in C_c^\infty(M^D),$$

namely a smooth compactly supported complex-valued function on M^D .

Consider a point-like quantum field

$$\hat{\Phi}_A(x),$$

which is an operator-valued distribution.

Its operator expression is the smeared field,

$$\boxed{\hat{\Phi}_A(f) = \int_{M^D} d^D x f(x) \hat{\Phi}_A(x),} \quad f \in C_c^\infty(M^D).$$

The field is always evaluated against a testing function.

The smeared field

$$\hat{\Phi}_A(f)$$

measures the field inside the spacetime region selected by f .

The integral sign is distributional notation for evaluating the field on the test function.

Equivalently,

$$\boxed{\hat{\Phi}_A(f) = \langle \hat{\Phi}_A, f \rangle.}$$

If

$$\hat{\Phi}_A(f)$$

is an unbounded operator,
then it requires a common dense domain.
Unbounded operators cannot act on the whole Hilbert space.

4 Unbounded Operators

Unbounded operators can only act on a dense domain

$$D \subset \mathcal{H}.$$

Therefore we cannot simply write

$$\hat{\Phi}(f) : \mathcal{H} \rightarrow \mathcal{H}.$$

Smearred fields are better behaved than point fields, but are still unbounded operators.
Hence one must specify their common domain.

4.1 $B(\mathcal{H})$ and $U(\mathcal{H})$

The algebra of bounded operators is

$$B(\mathcal{H}) = \{\text{bounded linear operators on } \mathcal{H}\}.$$

The unitary group is

$$U(\mathcal{H}) = \{U : \mathcal{H} \rightarrow \mathcal{H} \mid U^\dagger U = U U^\dagger = I\}.$$

It is the group formed by all unitary operators on the Hilbert space.

4.2 Hat, Unhatted, Tilde

- Hat:

$$\hat{\phi}, \quad \hat{\Phi}_A,$$

denote operator-valued distributions on Hilbert space.

- Unhatted:

$$\phi, \quad \psi, \quad \Phi_E,$$

denote classical fields, path-integral variables, Euclidean random fields, etc.

Example:

$$Z = \int D\phi e^{iS[\phi]}.$$

Here ϕ is an integration variable.

- Tilde:

denotes Fourier transform.

$$\tilde{\phi}(k) = \int d^D x e^{-ikx} \phi(x).$$

5 Test Function Space

Locality statements are formulated using

$$C_c^\infty(M^D).$$

Compact support controls localization inside a bounded spacetime region.

Suppose

$$f, g \in C_c^\infty(M^D).$$

If the supports of f and g are spacelike separated, then

$$\boxed{[\hat{\phi}(f), \hat{\phi}(g)] = 0.}$$

Equivalently,

$$[\hat{\phi}(f), \hat{\phi}(g)] = 0, \quad \text{if } \text{supp}(f) \text{ and } \text{supp}(g) \text{ are spacelike separated.}$$

The support of a test function,

$$\text{supp}(f),$$

is the spacetime region where

$$f(x) \neq 0.$$

The smeared field

$$\hat{\Phi}(f) = \int d^D x f(x) \hat{\Phi}(x)$$

measures the field inside

$$\text{supp}(f).$$

Since

$$f \in C_c^\infty(M^D),$$

the support is compact, i.e. the field is only probed in a finite spacetime region.

Two spacetime regions are spacelike separated when

$$ds^2 = -dt^2 + d\mathbf{x}^2 > 0,$$

or equivalently, they are outside each other's light cone.

Microcausality

Spacelike separated operators should commute,

$$\boxed{[\hat{\Phi}(f), \hat{\Phi}(g)] = 0,}$$

whenever

$$\text{supp}(f) \quad \text{and} \quad \text{supp}(g)$$

are spacelike separated.

6 Schwartz Space

The spectral condition, Wightman axioms, and Euclidean reconstruction frequently use the Schwartz space

$\mathcal{S}$.

For

$$\mathbb{R}^D,$$

$$\mathcal{S}(\mathbb{R}^D)$$

denotes the space of rapidly decreasing smooth functions.

For Fourier transforms,

$$\boxed{\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}.}$$

The Fourier transform preserves Schwartz space.

6.1 Tempered Fields and Compact Support

Since

$$C_c^\infty(\mathbb{R}^D) \subset \mathcal{S}(\mathbb{R}^D),$$

a field defined on Schwartz space automatically acts on compactly supported test functions.

Hence

$$\hat{\Phi}_A : \mathcal{S} \longrightarrow \text{operators}$$

implies

$$\boxed{\hat{\Phi}_A : C_c^\infty \longrightarrow \text{operators}.}$$

If a distribution is tempered, then it admits an extension to Schwartz space.

Tempered distributions grow at most polynomially, while the testing functions decay rapidly.

6.2 Lorentz Group

The connected proper orthochronous Lorentz group is

$$SO^+(1, D-1).$$

The complete Lorentz group is

$$O(1, D - 1),$$

which satisfies

$$\Lambda^T \eta \Lambda = \eta.$$

It contains several disconnected components, including

- spatial reflection,
- time reversal,
- determinant -1 transformations,
- time-orientation reversing transformations.

The connected component is

$$SO^+(1, D - 1) \subset O(1, D - 1).$$

Its properties are

- Proper:

$$\det \Lambda = +1,$$
 preserving spatial orientation.
- Orthochronous:
 preserving time orientation.
- Connected:
 continuously connected to the identity.

6.3 Connected Poincaré Group

The connected Poincaré group is

$$P_+^\uparrow = \mathbb{R}^D \rtimes SO^+(1, D - 1).$$

Here

$$\mathbb{R}^D$$

represents spacetime translations.

The symbol

$$\rtimes$$

denotes the semidirect product, since Lorentz transformations act on translations.

A group element acts as

$$(a, \Lambda) : x \mapsto \Lambda x + a.$$

The group multiplication law is

$$\boxed{(a, \Lambda)(b, \Sigma) = (a + \Lambda b, \Lambda \Sigma).}$$

7 Vacuum Minkowski Local Quantum Theory

A vacuum Minkowski quantum field theory consists of

1. A complex Hilbert space

$$\mathcal{H}.$$

2. A spacetime symmetry structure.

Let

$$G_{\text{Lor}} = SO^+(1, D-1),$$

or more generally let

$$\pi : G_{\text{Lor}} \longrightarrow SO^+(1, D-1)$$

be the covering projection.

The connected Poincaré group is

$$P = \mathbb{R}^D \rtimes G_{\text{Lor}}.$$

Its group multiplication is

$$(a, \Lambda)(b, \Gamma) = (a + \pi(\Lambda)b, \Lambda\Gamma).$$

It acts on spacetime by

$$gx = \pi(\Lambda)x + a, \quad g = (a, \Lambda) \in P.$$

For a spacetime region

$$O \subset M^D,$$

define

$$gO = \{gx : x \in O\}.$$

3. A strongly continuous unitary representation

$$U : P \longrightarrow U(\mathcal{H}).$$

For every spacetime symmetry

$$g \in P,$$

there exists a unitary operator

$$U(g)$$

acting on the Hilbert space such that

$$\boxed{U(g_1 g_2) = U(g_1) U(g_2).}$$

The symmetry acts not only on spacetime points,

$$x \mapsto gx,$$

but also on quantum states,

$$|\psi\rangle \longmapsto U(g)|\psi\rangle.$$

If the spacetime symmetry changes continuously, then

$$U(g)|\psi\rangle$$

also changes continuously.

4. Translation generators and the spectrum condition.

For translations,

$$U(a, \mathbf{1}) = \exp(ia_\mu P^\mu) = \exp(ia^\mu P_\mu).$$

The momentum operator is

$$P^\mu = (P^0, P^1, \dots, P^{D-1}),$$

where

$$P^0 = H$$

is the Hamiltonian.

For a time translation,

$$\boxed{U(t) = e^{-iHt}.$$

The joint spectrum is

$$\text{supp}(E_P),$$

where

$$E_P$$

is the joint projection-valued spectral measure.

The spectrum condition requires

$$\boxed{\text{supp}(E_P) \subset \overline{V}_+,}$$

where

$$\overline{V}_+ = \{p \in \mathbb{R}^D : p^0 \geq 0, \eta_{\mu\nu} p^\mu p^\nu \leq 0\},$$

the closed future light cone.

Equivalently,

$$p^0 \geq |\mathbf{p}|, \quad p^0 \geq 0.$$

Massive particles satisfy

$$p^2 = -m^2 < 0,$$

while massless particles satisfy

$$p^2 = 0,$$

lying on the light cone.

5. Poincaré-invariant vacuum vector.

There exists a vacuum vector

$$\Omega \in \mathcal{H},$$

such that

$$\boxed{U(g)\Omega = \Omega, \quad \forall g \in P.}$$

The vacuum is unchanged under every Poincaré transformation.

Vacuum uniqueness:

$$E_P(\{0\})\mathcal{H} = \mathbb{C}\Omega,$$

where

$$E_P(\{0\})$$

is the projection onto the zero energy-momentum subspace.

If

$$P^\mu = 0,$$

then any state

$$|\psi\rangle$$

with

$$P^\mu |\psi\rangle = 0$$

must satisfy

$$\boxed{|\psi\rangle = c\Omega.}$$

Hence there is only one Poincaré-invariant vacuum.

6. Local observable algebras.

For every bounded open region

$$O \subset M^D,$$

there is an assignment

$$O \longmapsto \mathcal{A}(O) \subset B(\mathcal{H}),$$

where

$$B(\mathcal{H}) = \{\text{bounded operators on } \mathcal{H}\}.$$

If

$$\text{supp}(f) \subset O,$$

then

$$\Phi(f) \in \mathcal{A}(O).$$

The net of local algebras satisfies three fundamental properties.

(a) Covariance.

$$\boxed{U(g)\mathcal{A}(O)U(g)^{-1} = \mathcal{A}(gO).}$$

A Poincaré transformation maps observables in O to observables in the transformed region gO .

(b) Isotony.

If

$$O_1 \subset O_2,$$

then

$$\boxed{\mathcal{A}(O_1) \subset \mathcal{A}(O_2).}$$

(c) Locality (Microcausality).

If

$$O_1 \quad \text{and} \quad O_2$$

are spacelike separated,
then

$$\boxed{[A, B] = 0,}$$

for

$$A \in \mathcal{A}(O_1), \quad B \in \mathcal{A}(O_2).$$

Graded Commutator

If fermionic operators are present, one replaces the ordinary commutator by the graded commutator,

$$\boxed{[A, B]_{\text{gr}} = AB - (-1)^{|A||B|}BA.}$$

Here

$$|A|, |B| \in \mathbb{Z}/2\mathbb{Z}$$

denote the fermion parity (grading) of the operators.

If both operators are bosonic (even),

$$|A| = |B| = 0,$$

then

$$[A, B]_{\text{gr}} = [A, B] = AB - BA.$$

If both operators are fermionic (odd),

$$|A| = |B| = 1,$$

then

$$[A, B]_{\text{gr}} = AB + BA,$$

which is the anticommutator,

$$\boxed{\{A, B\} = AB + BA.}$$

7. Cyclicity of the vacuum (Reeh–Schlieder property).

The algebra generated by bounded local observables is denoted by

$$\mathcal{A}_{\text{loc}}^{\text{alg}}.$$

It is generated by taking all bounded local observables and closing under

- addition,
- multiplication,
- linear combinations,
- adjoints.

For example, starting from the local algebra

$$\mathcal{A}(O),$$

one generates

$$\mathcal{A}_{\text{loc}}^{\text{alg}}.$$

Every element

$$A \in \mathcal{A}_{\text{loc}}^{\text{alg}}$$

acts on the vacuum vector,

$$A\Omega.$$

Example (free scalar field):

$$\Phi(f)\Omega$$

generates a one-particle wave packet.

Repeated action,

$$\Phi(f_1) \cdots \Phi(f_n)\Omega,$$

generates multiparticle states.

The closure of these vectors satisfies

$$\overline{\{A\Omega : A \in \mathcal{A}_{\text{loc}}^{\text{alg}}\}} = \mathcal{H}.$$

8. Unbounded fields and field domain data.

Quantum fields are generally unbounded operators.

The point field

$$\Phi(x)$$

is an operator-valued distribution.

Hence the smeared field

$$\Phi(f)$$

is defined only on a common dense domain rather than on the whole Hilbert space.

Choose a collection

$$\left(D_{\Phi}, (\hat{\Phi}_A)_{A \in I} \right),$$

where

- I is the set of field labels (e.g. field components),

-

$$D_\Phi \subset \mathcal{H}$$

is a common dense domain,

-

$$\Omega \in D_\Phi,$$

- the domain is invariant under spacetime symmetry,

$$\boxed{U(g)D_\Phi \subset D_\Phi.}$$

Every smeared field is a linear operator

$$\boxed{\hat{\Phi}_A(f) : D_\Phi \longrightarrow D_\Phi,}$$

where

$$f \in T_A,$$

and T_A denotes the corresponding test-function space.